Trevor Chen

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Education

California Institute of Technology — Pasadena, CA

B.S. in Applied and Computational Mathematics | GPA: 4.0/4.0 — *Expected Jun. 2029*

- **Relevant Coursework:** Agentic Systems, Diffusion Models, Generative Models, Deep Learning, ML & Data Mining, GPU Programming (CUDA), Operating Systems, Statistical Inference, Probability Models, Applied Linear Algebra
- **Honors:** Y Combinator F25 Acceptance; YC AI Startup School; USACO Platinum; Regeneron STS Scholar; SPIE Scholarship; AIME Qualifier; National Merit Scholar; International Logic Olympiad Silver; Outstanding Presentation Award, Dayton-Cincinnati Aerospace Sciences Conference

Experience

Ascade (Accepted to Y Combinator F25) — Jun. 2025 – Present

Founder & CTO — *San Diego, CA*

- Shipped a formally verified payment processor to production, growing it to **$25K MRR**
- Won **DARPA ERIS** and federal contracts for formally verified zero-trust access control (IAM) for **AI agents**
- Trained an 8B **LLM** that sets **SOTA on formal-verification benchmarks**, beating frontier models (Claude Opus 4.6) at a fraction of the cost

Leonardo DRS Daylight Solutions — Jun. 2025 – Aug. 2025

Machine Learning Engineer — *San Diego, CA*

- Built and **deployed to production** an ML signal-processing model in MIRcat laser spectroscopy software, cutting water-vapor noise by **98%+** and improving SNR by **99%**
- Made it **50%** faster than legacy software through algorithmic optimization

Spectral Energies — Feb. 2022 – Jul. 2025

R&D Intern — *Remote*

- Designed broadband **mid-IR diffraction gratings** for high-energy ultrafast lasers via global optimization (**>90%** efficiency over 3.6–6.6 µm); prototypes hit **92%** vs. 94.2% theoretical
- Co-developed a **nanofabrication** process with Penn State Nanofab (**electron-beam lithography** + **anisotropic wet etching**) for near-perfect blazed grooves, improving efficiency by **30%**
- Optimized hybrid metal–dielectric **thin-film coatings** (Au–Al_2O_3) to raise efficiency and **laser damage threshold**
- Discovered a "double-blazing" grating design in simulation (**+50%** bandwidth, **+20%** efficiency); simulation pipelines cut prototyping cycles by **70%** and manufacturing error by **80%**
- Delivered 10 gratings deployed at 5+ national labs, supporting **$1.3M+** in R&D contract revenue

Research Experience

ETH Zurich AI Center — Krause Group (joint with Arnold Lab, Caltech) — Oct. 2026 – Present

Researcher, with Prof. Andreas Krause and Prof. Frances Arnold — *Remote*

- **AI for Science:** building a **closed-loop discovery system** where a frozen **LLM** proposes mechanistic hypotheses, each distilled into a domain-specific **flow model** that generates testable designs, with **wet-lab** outcomes certifying or refuting each hypothesis
- Developing signed distillation that folds validated hypotheses into the generator and suppresses refuted ones, anchored to the previous model to avoid collapse, for campaigns limited to $\sim 10^2$ wet-lab measurements
- Designing the evaluation against active search, reward-based fine-tuning, and **agentic LLM** loops across **enzyme** and **protein design** tasks, with wet-lab validation in Frances Arnold's lab

Yue Lab, Caltech — Jan. 2026 – Present

Undergraduate Researcher, advised by Prof. Yisong Yue — *Pasadena, CA*

- **AI for Drug Discovery (with Genesis Molecular AI):** led an 11-author study building a **sample-efficient reinforcement learning** and **active learning** framework for **diffusion models**, with **58**× fewer evaluations than the best baseline (**NeurIPS 2026**)
- **Generative Model Theory:** showed why larger latent spaces delay **memorization** in diffusion models (**3.6**× later), explaining **generalization** through **training dynamics** (**ICML 2026 HiLD**; **ICLR 2027** under review)
- Scaled **distributed training** across multi-node **GPU** clusters (HPC, Slurm) with fault-tolerant scheduling

Caltech SURF Fellow — Perona Vision Lab — Jun. 2026 – Aug. 2026

Summer Undergraduate Research Fellow, advised by Prof. Pietro Perona — *Pasadena, CA*

- **SAE methods:** discovered a family of **sparse autoencoder (SAE)** training objectives that steer which features a model learns, so researchers can target the features they want instead of relying on random seeds (**CVPR 2027**, under review)
- **Feature evaluation:** showed the steered features are functionally different, not cosmetic, with **linear probes** and planted-signal recovery

- **Interpreting generative models:** trained a **diffusion transformer** from scratch plus 36 SAEs (**220K+** features) across every layer's residual stream and attention/MLP outputs
- **Causal analysis:** ran **feature-ablation interventions** across the denoising process to trace how features drive generation

UCSD San Diego Supercomputer Center Oct. 2020 – May 2025
Computational Biology Research Intern *San Diego, CA*

- Built ML models for **AI-driven drug discovery** (protein–ligand binding prediction) at **87%** accuracy
- Ran large-scale **virtual screening** of **50K+** compounds on **HPC** clusters for HIV and diabetic-cataract drugs
- Improved binding prediction accuracy by **20%** with simulation-based **data augmentation** (3 papers)
- Mentored 3 interns, teaching molecular docking simulations and helping validate their ML models

Projects

Breaking Pangram's AI-Image Detector | *Independent Red-Teaming* Aug. 2026

- **AI Safety / Red-Teaming:** broke a leading commercial AI-content detector **in both directions** — made AI images pass as human and real photos get flagged as AI — using **adversarial training** of generative models (VAE, GAN)
- Exposed false positives on upscaled real photos; disclosed findings to Pangram's co-founder; withheld the attack for safety

Real-Time CUDA Black Hole Renderer | *CS 179 GPU Programming* Jun. 2026

- Co-built a real-time **GPU ray tracer** in **CUDA/C++** that bends light through curved spacetime per pixel, rendering gravitational lensing, Einstein rings, and a Doppler-shifted accretion disk
- Wrote the **CUDA** rendering and post-processing kernels (bloom, HDR tone mapping) and an interactive browser viewer with live controls and FPS profiling
- Validated against general relativity with a 12-test physics suite (light deflection within **0.44%** of theory, energy drift $<\mathbf{10}^{-6}$) plus CPU–GPU parity tests

Medium-Frequency Trading on Binary Prediction Markets | *Independent Project* May 2025 – Aug. 2025

- Built an **algorithmic trading** system for prediction markets with **reinforcement learning** and **NLP**, cutting slippage **30%**
- Ran arbitrage and hedging strategies for consistent **8%** weekly returns over 3 months via a **6**× faster execution API

Skills

Languages: Python, C++, C, CUDA, SQL, Lean 4, Dafny
ML / AI: PyTorch, NumPy, Deep Learning, Generative AI, LLMs, Transformers, Diffusion Models, Reinforcement Learning, Fine-Tuning, Active Learning, Evals & Benchmarking
Interpretability & Safety: Sparse Autoencoders (TopK), Dictionary Learning, Causal Interventions & Feature Ablation, Linear Probing, Representation Analysis, Red-Teaming, Adversarial Robustness
Infrastructure: Distributed Training, Multi-GPU, HPC, Slurm, Linux, Git
AI for Science: Boltz, ESM3, GenMol, DISCO, Drug Discovery, Protein Engineering, Molecular Optimization
Other: Formal Verification, AI Agents, Nanofabrication, Optical Simulation

Publications & Invited Talks

[*] *Equal contribution (shared first author)*

T. Chen[*], A. Dai[*], J. Yang, R. De Santi, *et al.*, M. Al-Shedivat, Y. Yue. *On the Design Space of Discrete Diffusion Online Adaptation for Molecular Optimization*. **NeurIPS 2026**.

T. Chen, R. Shahbaba. *How Excess Latent Dimensionality Delays Memorization in Diffusion Models*. **ICLR 2027** (under review).

T. Chen, R. Shahbaba, K. Peng, A. Garg. *How Excess Latent Dimensionality Delays Memorization in Diffusion Models*. **ICML 2026 Workshop on High-dimensional Learning Dynamics (HiLD)**.

T. Chen, I. Demler, P. Perona. *Objective Geometry Steers Sparse Autoencoder Vocabularies*. **CVPR 2027** (under review).

J. Y. Tang[*], **T. Chen**[*], V. L. Kouznetsova, I. F. Tsigelny. *Anelloviruses and Cancer*. **The Journal of Infectious Diseases** 231(2), 2025.

T. Chen, B. Liu, J. A. McCoy, *et al. Development of Broadband High-Efficiency Mid-IR Gratings for High-Energy Ultrafast Lasers*. **Optical Materials Express** 14(5), 2024.

T. Chen, P. S. Hsu. *Enabling Ultra-Broadband High-Efficiency Gratings for Single-Cycle Infrared Lasers*. **SPIE Nanoengineering**, 2024.

P. S. Hsu[*], **T. Chen**[*], A. Roy, *et al. A Compact LWIR Borescope Sensor for 2D Engine Component Surface Temperature Measurement*. **Measurement Science and Technology** 34(6), 2023.

A. Roy, P. S. Hsu, A. Zhang, **T. Chen**, *et al. Development of Compact LWIR Borescope Sensor for Non-contact 2D Engine Component Surface Temperature Measurement*. **AIAA AVIATION Forum**, 2023.

T. Chen, A. Zhang, A. Roy, P. S. Hsu, O. Bibik, S. Adhikari. *2D High Temperature Sensor Development Using LWIR Borescope*. **SPIE Infrared Sensors, Devices, and Applications XIII**, 2023.

T. Chen, R. Chen, A. You, V. L. Kouznetsova, I. F. Tsigelny. *Search of Inhibitors of Aldose Reductase for Treatment of Diabetic Cataracts Using Machine Learning*. **Advances in Ophthalmology Practice and Research** 3(4), 2023.

A. K. Gao, **T. Chen**, V. L. Kouznetsova, I. F. Tsigelny. *Machine-Learning-Based Virtual Screening and Ligand Docking Identify Potent HIV-1 Protease Inhibitors*. **Artificial Intelligence Chemistry** 1(2), 2023.

T. Chen, P. S. Hsu. *Compact Borescope Design for 2D Long-Wavelength IR Thermal Imaging Sensing*. **Optica LACSEA**, 2022.

Invited Talks

Invited Ethics Speaker, iGEM Conference.
Invited Researcher, Leonardo DRS Daylight Solutions Executive Team.